

CHRIS D'SOUZA

PCCE

Frontend:

```
ng serve -o (signup-frontend) X + v
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP>cd C:\Users\HP\Downloads

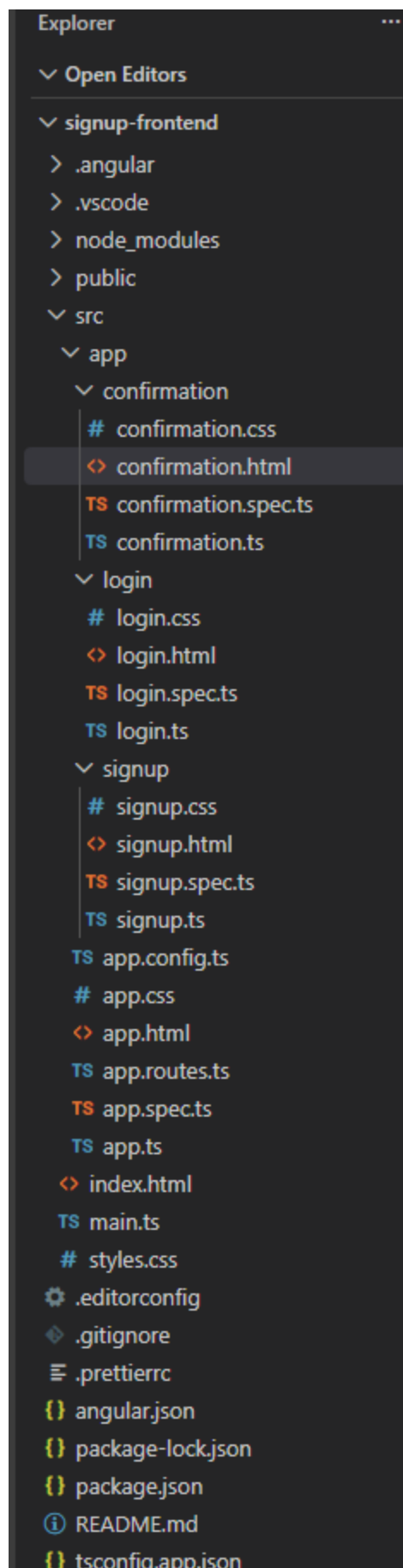
C:\Users\HP\Downloads>ng new signup-frontend --routing=true --style=css
✓ Do you want to enable Server-Side Rendering (SSR) and Static Site Generation (SSG/Prerendering)? No
✓ Which AI tools should Angular integrate with? https://angular.dev/ai/develop-with-ai None
CREATE signup-frontend/.prettierrc (173 bytes)
CREATE signup-frontend/angular.json (2003 bytes)
CREATE signup-frontend/package.json (791 bytes)
CREATE signup-frontend/README.md (1526 bytes)
CREATE signup-frontend/tsconfig.json (939 bytes)
CREATE signup-frontend/.editorconfig (331 bytes)
CREATE signup-frontend/.gitignore (666 bytes)
CREATE signup-frontend/tsconfig.app.json (412 bytes)
CREATE signup-frontend/tsconfig.spec.json (423 bytes)
CREATE signup-frontend/.vscode/extensions.json (134 bytes)
CREATE signup-frontend/.vscode/launch.json (490 bytes)
CREATE signup-frontend/.vscode/tasks.json (1020 bytes)
CREATE signup-frontend/src/main.ts (228 bytes)
CREATE signup-frontend/src/index.html (313 bytes)
CREATE signup-frontend/src/styles.css (81 bytes)
CREATE signup-frontend/src/app/app.spec.ts (713 bytes)
CREATE signup-frontend/src/app/app.ts (310 bytes)
CREATE signup-frontend/src/app/app.css (0 bytes)
CREATE signup-frontend/src/app/app.html (20488 bytes)
CREATE signup-frontend/src/app/app.config.ts (322 bytes)
CREATE signup-frontend/src/app/app.routes.ts (80 bytes)
CREATE signup-frontend/public/favicon.ico (15086 bytes)
✓ Packages installed successfully.
warning: in the working copy of 'package-lock.json', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'package.json', LF will be replaced by CRLF the next time Git touches it
Successfully initialized git.

C:\Users\HP\Downloads>cd signup-frontend

C:\Users\HP\Downloads\signup-frontend>ng generate component login
CREATE src/app/login/login.spec.ts (547 bytes)
CREATE src/app/login/login.ts (191 bytes)
CREATE src/app/login/login.css (0 bytes)
CREATE src/app/login/login.html (21 bytes)

C:\Users\HP\Downloads\signup-frontend>ng generate component signup
CREATE src/app/signup/signup.spec.ts (554 bytes)
CREATE src/app/signup/signup.ts (195 bytes)
CREATE src/app/signup/signup.css (0 bytes)
CREATE src/app/signup/signup.html (22 bytes)

C:\Users\HP\Downloads\signup-frontend>ng generate component confirmation
CREATE src/app/confirmation/confirmation.spec.ts (596 bytes)
CREATE src/app/confirmation/confirmation.ts (219 bytes)
CREATE src/app/confirmation/confirmation.css (0 bytes)
CREATE src/app/confirmation/confirmation.html (28 bytes)
```



TS signup.ts

src > app > signup > TS signup.ts

```
1 import { Component } from '@angular/core';
2 import { CommonModule } from '@angular/common';
3 import { ReactiveFormsModule, FormBuilder, FormGroup, Validators, AbstractControl, ValidationErrors } from '@angular/forms';
4 import { HttpClient } from '@angular/common/http';
5 import { Router, RouterModule } from '@angular/router';
6
7 @Component({
8   selector: 'app-signup',
9   standalone: true,
10  imports: [CommonModule, ReactiveFormsModule, RouterModule],
11  templateUrl: './signup.html'
12 })
13 export class Signup {
14   signupForm: FormGroup;
15
16   constructor(private fb: FormBuilder, private http: HttpClient, private router: Router) {
17     this.signupForm = this.fb.group({
18       firstName: ['', [Validators.required, Validators.pattern('[a-zA-Z]+$')]],
19       lastName: ['', [Validators.required, Validators.pattern('[a-zA-Z]+$')]],
20       email: ['', [Validators.required, Validators.email]],
21
22       phone: ['', [Validators.required, Validators.pattern('^(:\\+?[1-9]\\d{12}|[1-9]\\d{9})$')]],
23
24       password: ['', [Validators.required, Validators.pattern('^(?=.*[a-zA-Z])(?=.*\\d)(?=.*[@$!%*?&])[6,]$')]],
25       repeatPassword: ['', Validators.required],
26       pincode: ['', [Validators.required, Validators.minLength(4)]]
27     }, { validators: this.passwordMatchValidator });
28   }
29
30   passwordMatchValidator(control: AbstractControl): ValidationErrors | null {
31     const password = control.get('password')?.value;
32     const repeatPassword = control.get('repeatPassword')?.value;
33     return password === repeatPassword ? null : { passwordMismatch: true };
34   }
35
36   onSubmit() {
37     if (this.signupForm.valid) {
38
39       this.http.post<any>('http://localhost:8080/api/auth/signup', this.signupForm.value)
40         .subscribe({
41           next: (response) => {
42             alert('Signup Successful!');
43             this.router.navigate(['/confirmation', response.id]);
44           },
45           error: (err) => {
46             alert('Error: ' + (err.error || 'Signup failed'));
47           }
48         });
49     } else {
50       alert('Please fill out the form correctly based on the validation rules.');
```

TS confirmation.ts

src > app > confirmation > TS confirmation.ts

```
1
2 import { Component, OnInit, ChangeDetectorRef } from '@angular/core';
3 import { CommonModule } from '@angular/common';
4 import { HttpClient } from '@angular/common/http';
5 import { ActivatedRoute, RouterModule } from '@angular/router';
6
7 @Component({
8   selector: 'app-confirmation',
9   standalone: true,
10  imports: [CommonModule, RouterModule],
11  templateUrl: './confirmation.html'
12 })
13 export class Confirmation implements OnInit {
14   userData: any;
15   errorMessage: string = '';
16
17   constructor(
18     private route: ActivatedRoute,
19     private http: HttpClient,
20     private cdr: ChangeDetectorRef
21   ) {}
22
23   ngOnInit() {
24     const id = this.route.snapshot.paramMap.get('id');
25
26     if (id) {
27       this.http.get<any>(`http://localhost:8080/api/users/${id}`).subscribe({
28         next: (data) => {
29           this.userData = data;
30
31           this.cdr.detectChanges();
32         },
33         error: (err) => {
34           this.errorMessage = 'Backend connection failed: ' + err.message;
35           this.cdr.detectChanges();
36         }
37       });
38     } else {
39       this.errorMessage = 'No user ID found in the URL.';
40     }
41   }
42 }
43
44 }
```

```

src > app > login > ts login.ts
1  import { Component } from '@angular/core';
2  import { CommonModule } from '@angular/common';
3  import { ReactiveFormsModule, FormBuilder, FormGroup, Validators } from '@angular/forms';
4  import { HttpClient } from '@angular/common/http';
5  import { Router, RouterModule } from '@angular/router';
6
7  @Component({
8      selector: 'app-login',
9      standalone: true,
10     imports: [CommonModule, ReactiveFormsModule, RouterModule],
11     templateUrl: './login.html'
12 })
13 export class Login {
14     loginForm: FormGroup;
15
16     constructor(private fb: FormBuilder, private http: HttpClient, private router: Router) {
17         this.loginForm = this.fb.group({
18             email: ['', [Validators.required, Validators.email]],
19             password: ['', Validators.required]
20         });
21     }
22
23     onSubmit() {
24         if (this.loginForm.valid) {
25             this.http.post<any>('http://localhost:8080/api/auth/login', this.loginForm.value)
26                 .subscribe({
27                     next: (response) => {
28                         alert('Login Successful!');
29
30                         this.router.navigate(['/confirmation', response.id]);
31                     },
32                     error: (err) => {
33                         alert('Invalid credentials. Please try again.');

```

ts app.routes.ts

```

src > app > ts app.routes.ts
1  import { Routes } from '@angular/router';
2
3
4  import { Login } from './login/login';
5  import { Signup } from './signup/signup';
6  import { Confirmation } from './confirmation/confirmation';
7
8  export const routes: Routes = [
9      { path: '', redirectTo: 'login', pathMatch: 'full' },
10     { path: 'login', component: Login },
11     { path: 'signup', component: Signup },
12     { path: 'confirmation/:id', component: Confirmation }
13 ];

```

```

C:\Users\HP\Downloads\signup-frontend>ng serve -o
Initial chunk files | Names | Raw size
main.js | main | 32.39 kB |
styles.css | styles | 95 bytes |

| Initial total | 32.48 kB

Application bundle generation complete. [1.732 seconds] - 2026-08-27T07:11:12.323Z

Watch mode enabled. Watching for file changes...
NOTE: Raw file sizes do not reflect development server per-request transformations.
→ Local: http://localhost:4200/
→ press h + enter to show help
Initial chunk files | Names | Raw size
main.js | main | 32.61 kB |

Application bundle generation complete. [0.255 seconds] - 2026-08-27T07:14:23.530Z

Page reload sent to client(s).

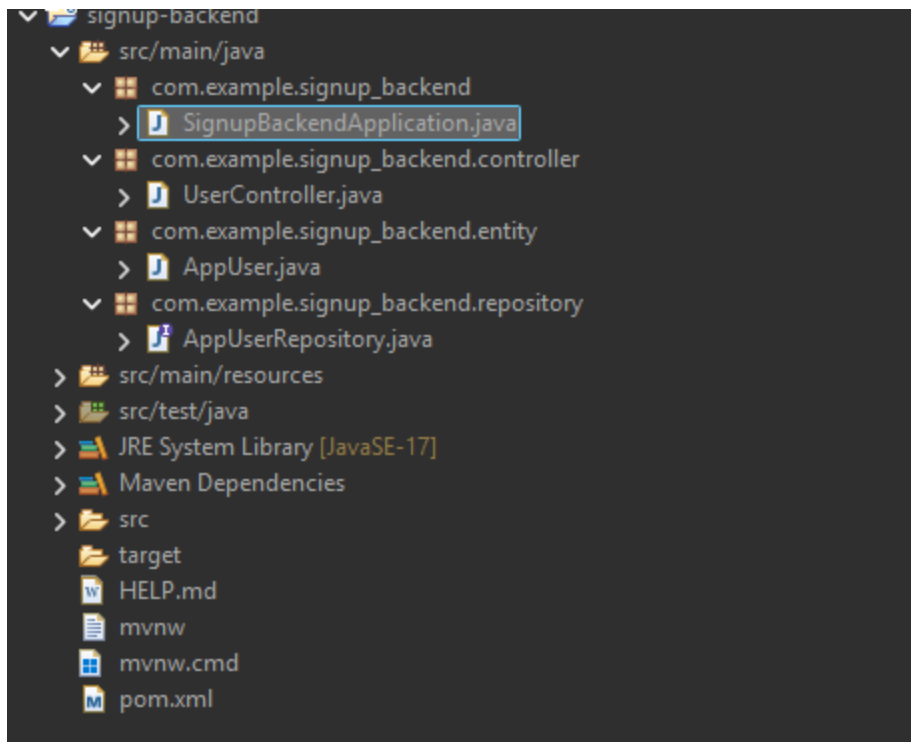
```

Backend:

```

<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
https://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <parent>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-parent</artifactId>
    <version>4.1.1</version>
    <relativePath>
      <!-- lookup parent from repository -->
    </relativePath>
  </parent>
  <groupId>com.example</groupId>
  <artifactId>signup-backend</artifactId>
  <version>0.0.1-SNAPSHOT</version>
  <name>
  <description>
  <url>
  <licenses>
    <license>
    </license>
  </licenses>
  <developers>
    <developer>
    </developer>
  </developers>
  <scm>
    <connection>
    <developerConnection>
    <tag>
    </tag>
    <url>
    </url>
  </scm>
  <properties>
    <java.version>17</java.version>
  </properties>
  <dependencies>
    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-data-jpa</artifactId>
    </dependency>
    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-webmvc</artifactId>
    </dependency>
    <dependency>
      <groupId>com.mysql</groupId>
      <artifactId>mysql-connector-j</artifactId>
      <scope>runtime</scope>
    </dependency>
    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-data-jpa-test</artifactId>
      <scope>test</scope>
    </dependency>
    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-webmvc-test</artifactId>
      <scope>test</scope>
    </dependency>
  </dependencies>
  <build>
    <plugins>
      <plugin>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-maven-plugin</artifactId>
      </plugin>
    </plugins>
  </build>
</project>

```



```
Problems Declaration Console X Eclipse IDE for Java Developers 2026-09-M3 Coverage
SignupBackendApplication [Java Application] C:\Program Files\Eclipse Adoptium\jdk-17.0.20-hotspot\bin\java.exe (27-Aug-2026, 12:11:42 pm elapsed 0:05:54) [pid: 20692]

Spring Boot (v4.1.1)

2026-08-27T12:11:44.018485:20 INFO 20692 --- [main] c.e.s.SignupBackendApplication : Starting SignupBackendApplication using Java 17.0.20 with PID 20692 (C:\Users\VP\Downloads\signup-backend\signup-backend\target\classes
2026-08-27T12:11:44.024485:30 INFO 20692 --- [main] c.e.s.SignupBackendApplication : No active profile set, falling back to 1 default profile: "default"
2026-08-27T12:11:44.668485:30 INFO 20692 --- [main] s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data JPA repositories in DEFAULT mode.
2026-08-27T12:11:44.734485:30 INFO 20692 --- [main] s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data repository scanning in 32 ms. Found 1 JPA repository interface.
2026-08-27T12:11:45.331485:30 INFO 20692 --- [main] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-08-27T12:11:45.346485:30 INFO 20692 --- [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-08-27T12:11:45.346485:30 INFO 20692 --- [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.24]
2026-08-27T12:11:45.488485:30 INFO 20692 --- [main] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 1319 ms
2026-08-27T12:11:45.689485:30 INFO 20692 --- [main] org.hibernate.orm.jpa : HH000054H: Processing PersistenceUnitInfo [name: default]
2026-08-27T12:11:45.695485:30 INFO 20692 --- [main] org.hibernate.orm.core : HH0000001: Hibernate ORM core version 7.4.5.Final
2026-08-27T12:11:45.135485:30 INFO 20692 --- [main] o.s.o.j.p.SpringPersistenceUnitInfo : No loadTimeWeaver setup; ignoring JPA class transformer
2026-08-27T12:11:46.136485:30 INFO 20692 --- [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Starting...
2026-08-27T12:11:46.364485:30 INFO 20692 --- [main] com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Added connection com.mysql.cj.jdbc.ConnectionImpl@7879f8f0
2026-08-27T12:11:46.367485:30 INFO 20692 --- [main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
2026-08-27T12:11:46.413485:30 WARN 20692 --- [main] org.hibernate.orm.deprecation : HH00000025: MySQLDialect does not need to be specified explicitly using 'hibernate.dialect' (remove the property setting and it will be
2026-08-27T12:11:46.636485:30 INFO 20692 --- [main] org.hibernate.orm.connections.pooling : HH010061005: Database info:
Database JDBC URL [jdbc:mysql://localhost:3306/signup_db?createDatabaseIfNotExist=true&useSSL=false&serverTimezone=UTC]
Database driver: MySQL Connector/J
Database dialect: MySQLDialect
Database version: 8.0.46
Default catalog/schema: signup_db/undefined
Autocommit mode: undefined/unknown
Isolation level: REPEATABLE_READ [default: REPEATABLE_READ]
JDBC fetch size: none
Pool: DataSourceConnectionProvider
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown
2026-08-27T12:11:47.197485:30 INFO 20692 --- [main] org.hibernate.orm.core : HH0000489: No JTA platform available (set 'hibernate.transaction.jta.platform' to enable JTA platform integration)
2026-08-27T12:11:47.250485:30 INFO 20692 --- [main] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit 'default'
2026-08-27T12:11:47.310485:30 INFO 20692 --- [main] o.s.d.j.r.query.QueryEnhancerFactories : Hibernate is in classpath. If applicable, HQL parser will be used.
2026-08-27T12:11:47.550485:30 WARN 20692 --- [main] jpaBaseConfiguration$JpaWebConfiguration : spring.jpa.open-in-view is enabled by default. Therefore, database queries may be performed during view rendering. Explicitly configure
2026-08-27T12:11:47.804485:30 INFO 20692 --- [main] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-08-27T12:11:47.824485:30 INFO 20692 --- [main] c.e.s.SignupBackendApplication : Started SignupBackendApplication in 4.288 seconds (process running for 4.62)
2026-08-27T12:17:49.132485:30 INFO 20692 --- [nio-8080-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-08-27T12:17:49.133485:30 INFO 20692 --- [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-08-27T12:17:49.135485:30 INFO 20692 --- [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms
Hibernate: select au_0.id,au_0.email,au_0.first_name,au_0.last_name,au_0.password,au_0.phone,au_0.pincod from users au_0 where au_0.id=?
Hibernate: select au_0.id,au_0.email,au_0.first_name,au_0.last_name,au_0.password,au_0.phone,au_0.pincod from users au_0 where au_0.id=?
```

```

UserController.java × AppUser.java AppUserRepository.java
1 package com.example.signup_backend.controller;
2
3 import com.example.signup_backend.entity.AppUser;
4
5 @RestController
6 @RequestMapping("/api")
7 @CrossOrigin(origins = "**")
8 public class UserController {
9
10     @Autowired
11     private AppUserRepository userRepository;
12
13     @PostMapping("/auth/signup")
14     public ResponseEntity<?> registerUser(@RequestBody AppUser user) {
15         if (userRepository.findByEmail(user.getEmail()).isPresent()) {
16             return ResponseEntity.badRequest().body("Email is already taken!");
17         }
18         AppUser savedUser = userRepository.save(user);
19         return ResponseEntity.ok(savedUser);
20     }
21
22     @PostMapping("/auth/login")
23     public ResponseEntity<?> loginUser(@RequestBody AppUser loginRequest) {
24         Optional<AppUser> user = userRepository.findByEmail(loginRequest.getEmail());
25         if (user.isPresent() && user.get().getPassword().equals(loginRequest.getPassword())) {
26             return ResponseEntity.ok(user.get());
27         }
28         return ResponseEntity.status(401).body("Invalid credentials");
29     }
30
31     @GetMapping("/users/{id}")
32     public ResponseEntity<?> getUserById(@PathVariable Long id) {
33         Optional<AppUser> user = userRepository.findById(id);
34         if (user.isPresent()) {
35             return ResponseEntity.ok(user.get());
36         }
37         return ResponseEntity.notFound().build();
38     }
39
40     @DeleteMapping("/users/{id}")
41     public ResponseEntity<?> deleteUser(@PathVariable Long id) {
42         if (userRepository.existsById(id)) {
43             userRepository.deleteById(id);
44             return ResponseEntity.ok("User deleted successfully!");
45         }
46         return ResponseEntity.notFound().build();
47     }
48 }

```

```

UserController.java AppUser.java × AppUserRepository.java
1 package com.example.signup_backend.entity;
2
3 import jakarta.persistence.*;
4
5 @Entity
6 @Table(name = "users")
7 public class AppUser {
8
9     @Id
10     @GeneratedValue(strategy = GenerationType.IDENTITY)
11     private Long id;
12
13     private String firstName;
14     private String lastName;
15
16     @Column(unique = true)
17     private String email;
18
19     private String phone;
20     private String password;
21     private String pincode;
22
23     public Long getId() { return id; }
24     public void setId(Long id) { this.id = id; }
25     public String getFirstName() { return firstName; }
26     public void setFirstName(String firstName) { this.firstName = firstName; }
27     public String getLastName() { return lastName; }
28     public void setLastName(String lastName) { this.lastName = lastName; }
29     public String getEmail() { return email; }
30     public void setEmail(String email) { this.email = email; }
31     public String getPhone() { return phone; }
32     public void setPhone(String phone) { this.phone = phone; }
33     public String getPassword() { return password; }
34     public void setPassword(String password) { this.password = password; }
35     public String getPincode() { return pincode; }
36     public void setPincode(String pincode) { this.pincode = pincode; }
37 }

```



```
AppUserRepository.java X
1 package com.example.signup_backend.repository;
2
3 import com.example.signup_backend.entity.AppUser;
4
5
6
7 public interface AppUserRepository extends JpaRepository<AppUser, Long> {
8     Optional<AppUser> findByEmail(String email);
9 }
```

SignupFrontend X +

localhost:4200/signup

Import favorites New folder mysanad.mynabor... mysanad.mynabor...

Sign Up

First Name
Joel

Last Name
DSouza

Email
joelgmail.com
Invalid email format.

Phone Number
9923
Invalid phone format. No starting 0.

Password

Min 6 chars, 1 letter, 1 number, 1 special (@#\$%!).

Repeat Password

Passwords do not match.

Pincode

Submit

[Already have an account? Login here.](#)

9923809954

Password

.....

Repeat Password

.....|

Passwords do not match.

Login

Email

joel@gmail.com

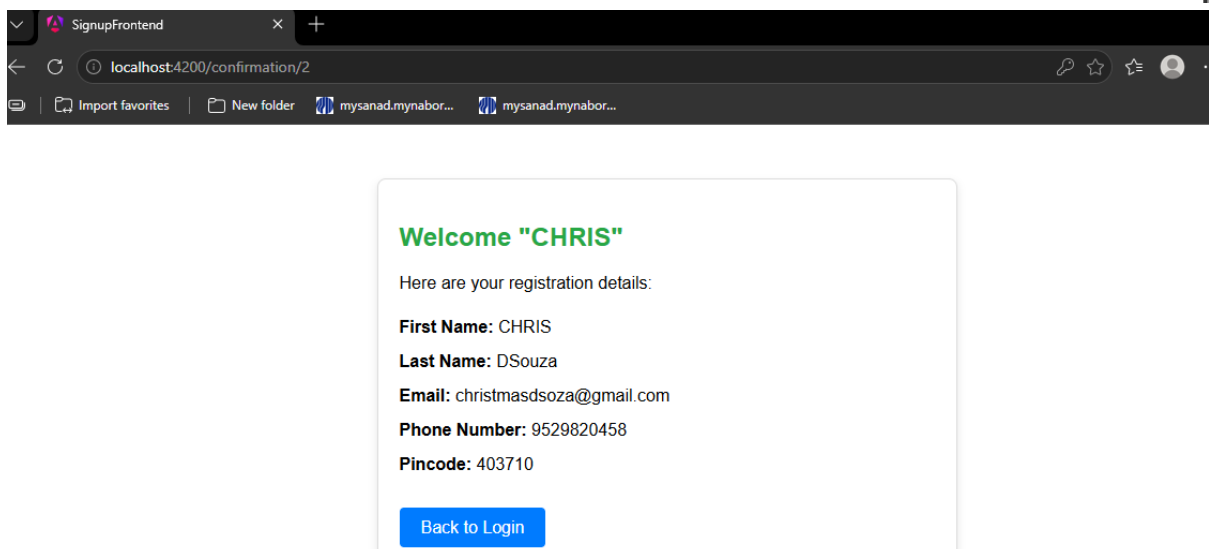
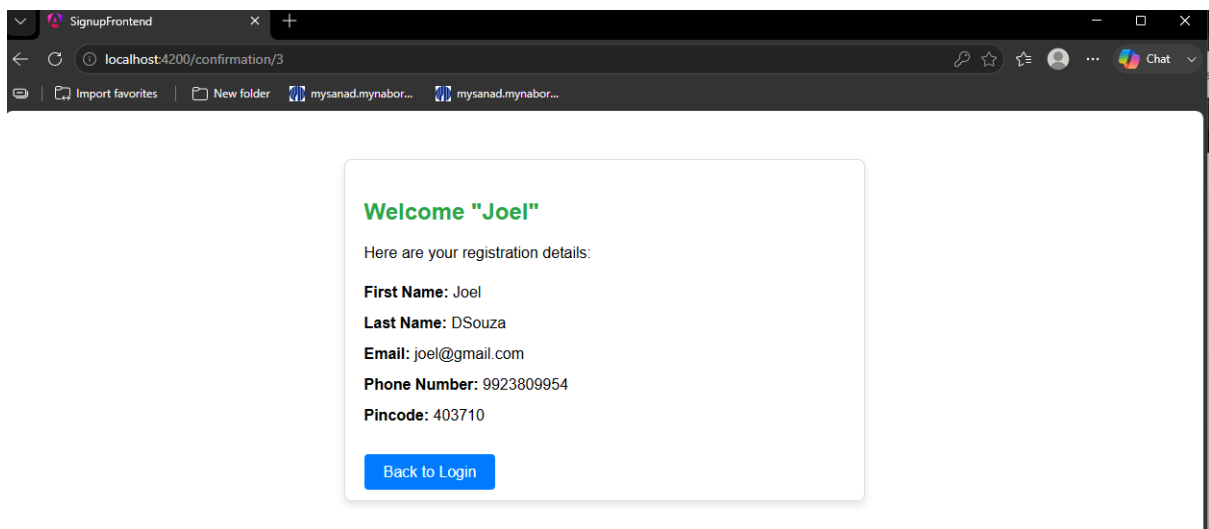
Password

.....



Login

[Don't have an account? Sign up here.](#)



OverviewFilesGET http://GET http://POST http://GET http://+No environment

HTTP

http://localhost:8080/api/users/2

SaveShare

GET

http://localhost:8080/api/users/2

Send

DocsParamsAuthorizationHeaders 6BodyScriptsSettings

Cookie

none

This request does not have a body

BodyCookiesHeaders 6Test Results

200 OK • 76 ms • 385 B •

{ } JSON

PreviewVisualize

```
1 {
2   "email": "christmasdsoza@gmail.com",
3   "firstName": "CHRIS",
4   "id": 2,
5   "lastName": "DSouza",
6   "password": "Qwerty@12",
7   "phone": "9529820458",
8   "pincode": "403710"
9 }
```

 **http://localhost:8080/api/users/3**  Save  Share

GET **Send**

Docs Params Authorization Headers 6 **Body** Scripts Settings

none

This request does not have a body

Body Cookies Headers 6 Test Results 

200 OK • 9 ms • 373 B •  ...

{ } JSON  Preview  Visualize

```
1 {
2   "email": "joel@gmail.com",
3   "firstName": "Joel",
4   "id": 3,
5   "lastName": "DSouza",
6   "password": "Astro@13",
7   "phone": "9923809954",
8   "pincode": "403710"
9 }
```

